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International Mathematics Assessments for Schools (2022-2023)

Junior Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Junior Division Round 2

Questions 1 to 5, 4 marks each

1. What is the total number of positive divisors of the expression
$$12\times11\times10\times9\times8\times7\times6\times5\times4\times3\times2\times1?$$
（A）5　（B）12　（C）100　（D）396　（E）792

Answer：_____

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2. Let a, b and c be real numbers such that $a+b+c=0$ and $abc=1$. What is the value of $a^3+b^3+c^3$?
（A）0　（B）1　（C）2　（D）3　（E）4

Answer：_____

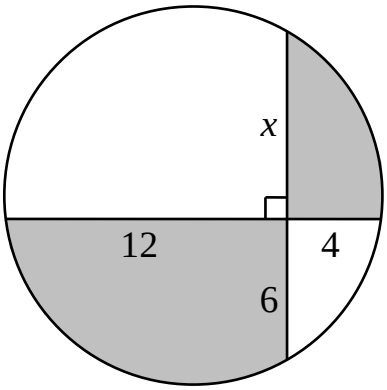
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3. A book contains 213 problems from 12 years of competitions. At first there were 25 problems per year, then it became 16 problems per year, and finally 20 per year. What was the number of years when there were 25 problems?
（A）1　（B） 2　（C）3　（D）5　（E）7

Answer：_____

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4. A circular pizza is sliced by using two perpendicular cuts, of lengths 16 cm and $x+6$ cm, as shown in the diagram below. Alice ate the two white pieces and Bob ate the two shaded pieces. Who ate more pizza? What is difference between the areas, in cm^2, of them?



（A）Alice, 16π cm^2　（B）Alice, 16 cm^2　（C）Equal amount, 0 cm^2
（D）Bob, 16 cm^2　（E）Bob, 16π cm^2

Answer：_____

5. In a certain class, more than 40%, but less than 45% of the students are girls. If one of the boys decided to leave the class, the percentage of girls would still be under 45%, but if a new girl joined the class, the percentage of girls would be higher that 45%. What is the smallest possible number of students in the class?
（A）10　　（B）20　　（C）30　　（D）40　　（E）50

Answer：_____

Questions 6 to 13, 5 marks each

6. What is the coefficient of x^2 in the product
$$(1-x)(1+2x)(1-2^2x)(1+2^3x)(1-2^4x)(1+2^5x)(1-2^6x)(1+2^7x)?$$

Answer：_____

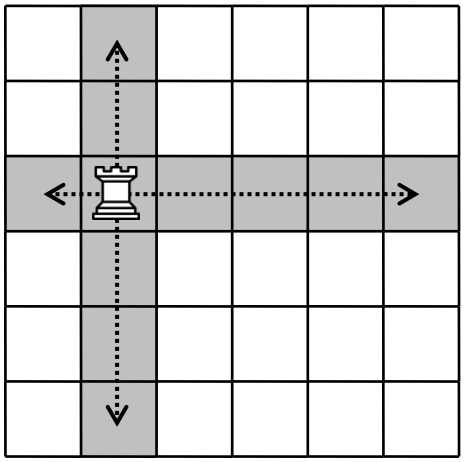
7. What is the minimum possible surface area, in cm^2, of a rectangular box having a volume strictly greater than 1000 cm^3 if each of the side lengths, in cm, are integer numbers?

Answer：_____ cm^2

8. Let A and B be disjoint subsets of $\{1, 2, \ldots, 100\}$ of equal size, where n belongs to A and $2n+2$ belongs to B. What is the maximum number of elements in $A \cup B$?

Answer：_____

9. Initially, there is a rook on each of the 36 squares of an 6×6 chessboard. In each move, one may take away any rook which attacks an odd number of other rooks still on the chessboard. What is the maximum number of rooks that can be removed? (Note: The Rook can move any number of spaces horizontally or vertically.)



Answer：_____ rooks

10. Determine the largest integer n such that $4^{27}+4^{100}+4^{n}$ is the square of an integer.

Answer : _____

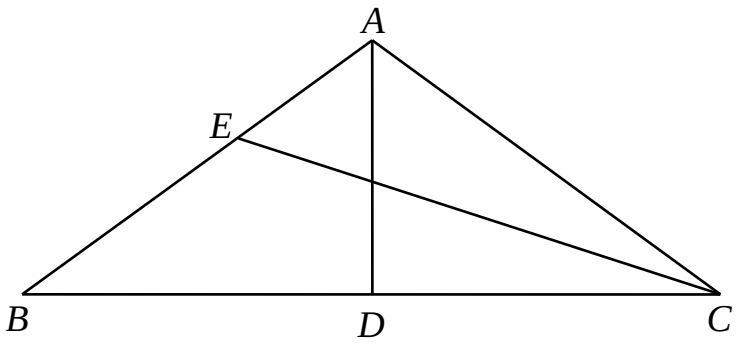
11. A small movie theater has 30 seats; their numbers are indicated on the tickets. The first 21 people who come to a show take 21 seats without paying attention to the seat numbers. Edmond is the 21st person to arrive. But the remaining 9 ticket holders are perfectionists. If any of them finds his or her assigned seat occupied, the person sitting there is evicted; that person then looks for his or her proper seat and evicts the occupier; and so forth. This migration ends with the spectator whose assigned seat is unoccupied. Find the probability, in percentage, that Edmond will not have to change his seat?

Answer : _____ %

12. Three students write on the blackboard three 2-digit squares next to each other. At the end they observe that the 6-digit number obtained is also a square. Find the sum of all possible 6-digit numbers.

Answer : _____

13. As shown in the figure below, triangle ABC is isosceles with $AB = AC$. It is given that D is the foot of the altitude from the vertex A and E lies on side AB such that $\angle ACE = \angle ECB = 18°$. If $AD = 24$ cm, find the length, in cm, of CE.



Answer : _____ cm

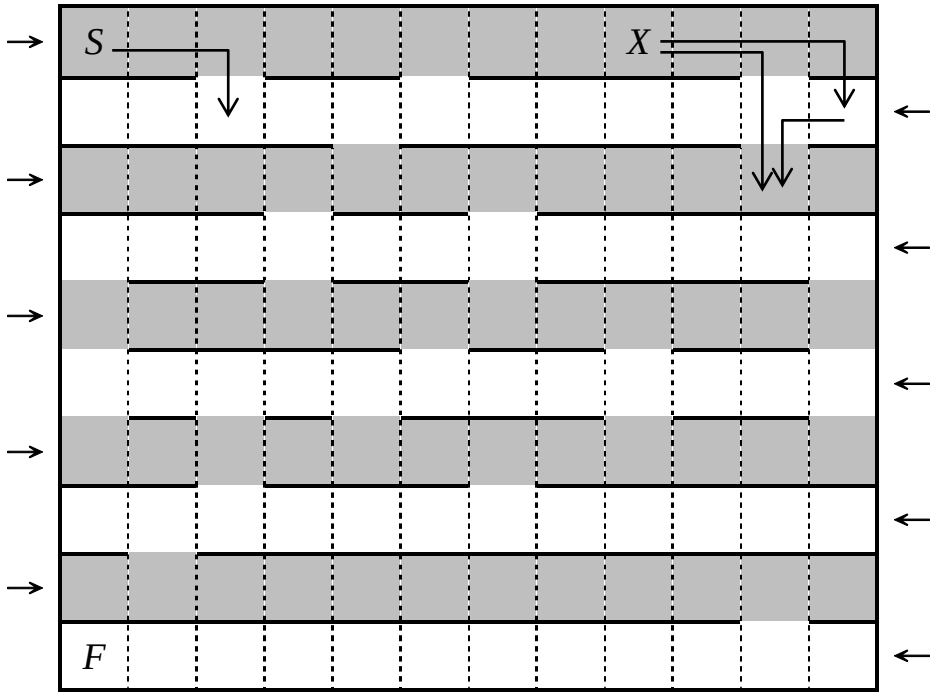
Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. A Mathematics test consists of 4 problems, where each problem is being graded independently with integer points between 0 and 10. How many ways can the total points for this test be exactly 31?

Answer： _____ ways

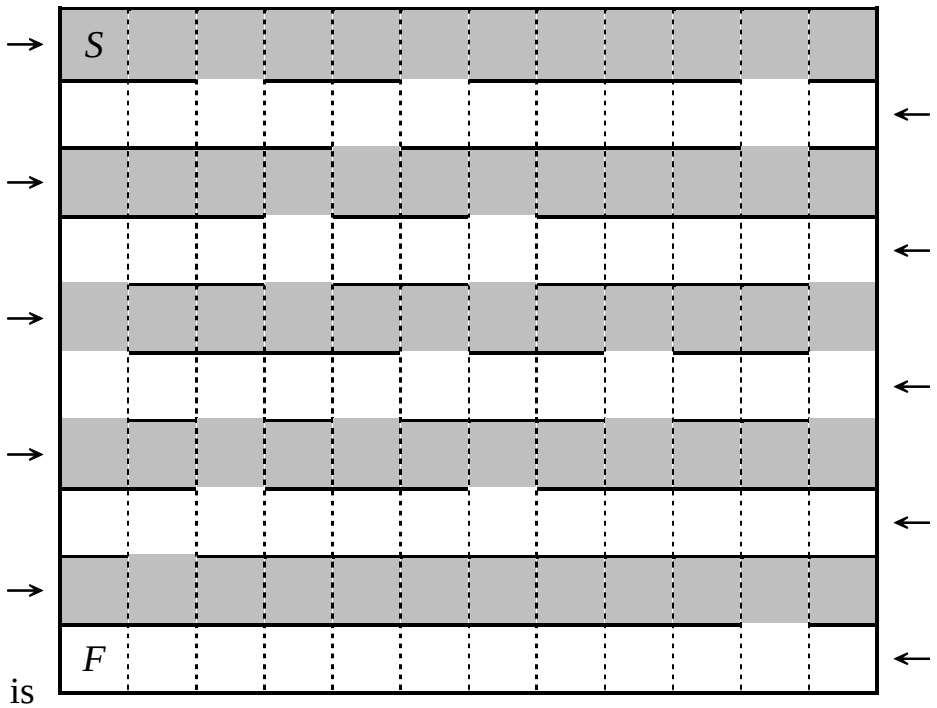
15. Andy has to pass through an underground maze as the diagram below. Each square means a room. Andy can pass through the dash segment to enter the adjacent room. If there is no segment between the adjacent rooms, then it means there is an open door between the adjacent rooms. Andy will move from 1 to 6 rooms by rolling a dice. The enchantment on the maze makes Andy drop one, two or three rows if he lands on a room with a door. For example, if Andy rolls a 2 from room *S*, then he would drop to the second row; if Andy rolls a 2 from room *X*, then he would drop to the third row.

    Andy is required to move from left to right on the odd row and from right to left on the even row. At the end of a row, the enchantment requires Andy to move down to the next row. For example, when Andy rolls a 5 from room *X*, he would enter the second row at fourth move and drop to the third row at the fifth move.



    As Andy arrive in room *F*, he will stop. What is the minimum number of dice rolls required to let Andy move from room *S* to room *F*? Please explain in detail your reasoning and draw the route map.

Answer： The fewest number of rolls of dice is _____ .



The route map is _____